

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Forth Semester of B. Tech. (CE/IT) Examination

April 2013

IT206.01 Object Oriented Programming using Java

Date: 23.04.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Q-1 Do as directed.

07

1. Which of the following does not denote a primitive data value in Java?
 - (a) "t"
 - (b) 'k'
 - (c) 50.5F
 - (d) false
2. The _____ reference to the current object is useful in situations where a local variable hides, or *shadows*, a field with the same name.
 - (a) this
 - (b) super
 - (c) private
 - (d) None of the above
3. If a class defines any explicit constructors, it can no longer rely on the implicit default constructor to set the state of its objects. **State True/False. Justify your answer.**
4. A package hierarchy represents an organization of the Java _____ and interfaces.
 - (a) Classes
 - (b) Inheritances
 - (c) Exceptions
 - (d) I/O streams
5. Static code is executed in the context of an object, therefore the references this and super are available for it. **State True/False. Justify your answer.**
6. If a class that has no abstract methods, it may be declared abstract. **State True/False. Justify your answer.**
7. A private member is accessible in all classes in the same package and by all subclasses of its class in any package where that class is visible. **State True/False. Justify your answer.**

Q-2

- (a) What are the purposes of this and super in Java? Why does *this()* and *super()* have to be the first statement in a constructor? 04
- (b) Can an abstract class be final? Why? Can abstract class have a constructor? Justify your answer. 05
- (c) Write a Java program for Given two arrays e.g., 1, 2, 3, 4, 5 and 2, 3, 1, 0, 5 to find which number is not present in the second array. 05

OR

- Q-2**
- (a) Discuss the working and meaning of the “static” modifier with example. Write the differences between class variable and instance variable. 04
 - (b) How objects are constructed? Explain constructor overloading with an example. Why constructor doesn't have return type? 05
 - (c) Write a program to determine the sum of the series for a given value of n. 05
 $1 + 1/2 + 1/3 + \dots 1/n$

- Q-3**
- (a) What is a Package? What are the benefits of using packages? Write down the steps in creating a package and using it in a java program with an example. 06
 - (b) Design a Java interface for numbers. Develop two different classes that implement this interface, one for to check whether the given number is palindrome or not and the other to check whether the given number is odd or even. 08

OR

- Q-3**
- (a) What is an Exception? What is the difference between checked exception and unchecked exception? Explain how to throws and try catch handle the Exceptions. 06
 - (b) Why java has Thread class and Runnable interface for implementation of thread? Write a java program that prints numbers from 1 to 10 line by line after every 5 seconds. 08

Section II

- Q-4**
- (a) Give the difference between byte oriented stream and character oriented stream. List the basic classes of both the streams. 03
 - (b) List and explain the life cycle methods of Applet. Can applet have a main method? Justify your answer. 04

- Q-5**
- (a) Discuss the overview of Object Oriented Analysis. Describe the basic activities in Object Oriented Analysis and explain how Use-case Modeling is useful in analysis. 06
 - (b) Discuss the significance of Unified Approach. Define UML Diagrams. List out the different types of UML Diagrams. Explain in brief about all the UML diagrams that represent only the dynamic aspects of the software system with example. 08

OR

- Q-5**
- (a) What is Generalization? Explain three purpose of Generalization in Object Oriented Development with example. 06
 - (b) Draw the use case diagram and class diagram for Railway Reservation System. 08

- Q-6**
- (a) Difference between following: 06
 1. State Chart Diagram and Activity Diagram.
 2. Aggregation and Composition
 - (b) List and Explain the basic four elements of component diagram with example. Also Differentiate between component diagram and deployment diagram. 08

OR

- (b) What are the similarities/dissimilarities between a sequence diagram and collaboration diagram? Explain with suitable example covering all the aspects. Prepare sequence diagram for withdrawal of money from Automated Tailor Machine (ATM) for all three scenarios. 08